

Strategic Blueprint and Technical Architecture for the RoadSOS Emergency Response System: National Road Safety Hackathon 2026

The Imperative for Data-Driven Mobility Governance

The intersection of artificial intelligence and civic infrastructure represents a critical frontier in modern public administration and public health policy. The National Road Safety Hackathon 2026, orchestrated by the Centre of Excellence for Road Safety (CoERS) at RBG Labs, Indian Institute of Technology (IIT) Madras, seeks to harness this intersection to address a profound national and global crisis.¹ With the Republic of India experiencing an estimated 1.7 lakh annual road fatalities, the socio-economic toll is staggering, accounting for an estimated societal cost equivalent to three percent of the national Gross Domestic Product.¹ Traditional, reactive enforcement models and isolated emergency response units have proven systematically insufficient to curb this epidemic.⁴ Consequently, the overarching theme of the 2026 hackathon is defined as “AI in Road Safety,” establishing a mandate for the creation of scalable, evidence-based, and implementation-ready innovations that can be seamlessly integrated into real-world, decentralized governance frameworks.²

The institutional context surrounding this initiative is of paramount importance for any systems architect aiming to develop a viable solution. CoERS operates under the direct aegis and funding of the Ministry of Road Transport and Highways (MoRTH), Government of India, and applies an evidence-based, human-centric “5E” Model for Road Safety: Engineering, Enforcement, Emergency Care, Education, and Empathy.¹ The institution has already deployed a sophisticated suite of AI-enabled governance platforms that form the digital backbone of India's evolving road safety infrastructure. Notable among these are ‘Sanjaya,’ a national crash intelligence and location analytics tool currently deployed in states such as Haryana and Odisha, which provides single-platform visualization of crash black spots through heatmaps and corridor analysis.⁷ Furthermore, the ‘RATH’ platform serves as an AI-enabled data-driven governance dashboard for road safety stakeholders, while the Data-Driven Hyperlocal Interventions (DDHI) framework has been launched across numerous districts, including active deployments in New Delhi and Madhya Pradesh, to prioritize evidence-based interventions at high-risk locations.¹ Recently, CoERS introduced ‘ThinnAI,’ a personalized AI-enabled trainer designed to enhance driver readiness through multi-level, video game-like tests evaluating cognitive and physical readiness with zero human intervention.¹

Any competitive software solution developed for the 2026 hackathon must align conceptually and architecturally with this existing, sophisticated ecosystem. The mandate is not merely to build a standalone mobile application, but to architect a constituent node within a broader,

data-driven mobility governance network that supports the national vision of Viksit Bharat 2047.⁴ Participants are required to choose from one of three distinct problem statements: DriveLegal, an compliance tool integrating national and local traffic laws; RoadWatch, a civic transparency tool for monitoring public spending on road infrastructure; and RoadSOS.² This document provides an exhaustive technical blueprint, architectural specification, and strategic roadmap for tackling the RoadSOS problem statement, which is defined as an emergency response tool engineered to provide location-based access to nearby trauma centres, ambulance services, vehicle rescue units, police stations, and emergency contacts during road accidents.²

Hackathon Structural Guidelines and the Evaluative Matrix

To architect a winning solution for RoadSOS, the software development strategy must be rigorously reverse-engineered from the competition guidelines, the logistical framework, and the specific evaluation criteria utilized by the judging panel. Understanding the constraints and operational parameters of the hackathon is the foundational step in systems design.

The National Road Safety Hackathon 2026 is officially launched in tandem with the India AI Impact Summit 2026, positioning the initiative as a national priority within the broader artificial intelligence transformation agenda.¹ The competition process is structured across two distinct stages. The registration and code submission window commenced on March 11, 2026, and is scheduled to conclude on May 31, 2026.¹⁴ All registrations and project submissions must be channeled exclusively through the Unstop platform, a recognized portal for prestigious campus engagements.¹⁵ The eligibility criteria are intentionally broad to foster diverse technological perspectives: participation is open to all students, engineers, and problem solvers who are enthusiastic about ethics in technology, provided they meet the minimum age requirement of 15 years.² There are no mandatory professional qualifications, and the team composition rules are entirely unrestrictive, allowing individuals or groups of any size to compete.² The total financial prize pool includes ₹50,000 for the first prize, ₹30,000 for the second prize, and ₹20,000 for the third prize.² Beyond the monetary awards, all participants who successfully clear Stage 1 will be awarded a participation certificate, and selected candidates may be evaluated for highly coveted internship or job opportunities within the road safety technology sector.²

For Stage 1, the deliverable requirements dictate a highly specific, three-pronged submission protocol that tests both engineering prowess and the capacity for technical communication⁵:

1. **Comprehensive Source Code:** The complete, functional codebase must be submitted. Participants are granted the freedom to utilize any programming language or framework of their choice, provided the solution functions as intended.⁵
2. **Executive Presentation Deck:** A strictly enforced seven-slide presentation is required. This deck must include a "Welcome" and a "Thank You" slide, leaving exactly five slides to elucidate the solution's architecture, functionality, unique value proposition, and alignment with the MoRTH 5E model.⁵

3. **Technical Documentation:** A structured Word document must accompany the code, meticulously detailing all programmatic assumptions made during development and providing a comprehensive manifest of the software packages, libraries, Application Programming Interfaces (APIs), and frameworks utilized.⁵

The judging panel will not evaluate submissions based on subjective aesthetic appeal, but rather through a rigorous, predefined evaluative matrix specific to the RoadSOS track. Submissions will be scored across five primary dimensions, which must dictate the architectural priorities of the software.²

Evaluative Criterion	Architectural Translation	Strategic Focus Area
Reliability and Data Accuracy	Precision of geospatial algorithms and validity of fetched medical/police data.	Integration with verified government databases and dynamic algorithmic triage protocols.
Number of Contacts Fetched	Comprehensiveness of the spatial querying mechanism and database redundancy.	Utilizing OpenStreetMap alongside regional GIS platforms to ensure zero blind spots.
Offline Functionality	The system's capacity to operate in remote environments devoid of active internet.	Local SQLite databases, MBTiles vector mapping, and GSM modem SMS fallback systems.
Innovation & Additional Features	The integration of advanced computational models and automated hardware triggers.	Edge-based Artificial Intelligence for autonomous crash detection via smartphone sensors.
Information Integration Across Countries	Dynamic adaptability to cross-border jurisdictions and international protocols.	Geospatial geofencing engines that update emergency contact tables based on ISO country codes.

To satisfy these stringent criteria, the RoadSOS architecture must transcend the rudimentary functionality of a simple digital phone directory. It must operate as an autonomous,

context-aware, and highly resilient emergency response node. The proposed technical architecture comprises five interconnected subsystems: the Automated AI Crash Detection Module, the Geospatial Intelligence Module, the Resilient Offline-First Module, the Integrated Health and Trauma Triage Module, and the Global Protocol Resolution Module.

Automated AI Crash Detection Module: Sensor Fusion and Edge Inference

A critical innovation that elevates a standard SOS application into a proactive, life-saving instrument is the integration of Edge Artificial Intelligence for automated crash detection.¹⁷ The hackathon's mandate for "Innovation & Additional Features" is perfectly addressed by shifting the paradigm from reactive user input to proactive algorithmic intervention.² In severe road traffic accidents, victims frequently suffer from blunt force trauma, concussions, or traumatic brain injuries, rendering them incapacitated and physically unable to unlock a smartphone, open an application, and manually trigger an SOS alert.¹⁸ Therefore, RoadSOS must leverage smartphone sensor fusion to detect catastrophic kinematic events autonomously and trigger the emergency protocol without human intervention.¹⁷

Modern smartphones and wearable devices are universally equipped with sophisticated micro-electromechanical systems (MEMS), specifically three-axis accelerometers and three-axis gyroscopes, which continuously output spatial orientation and acceleration data at high sampling rates, typically around 50 Hz.¹⁹ The fundamental physics of a vehicular crash or a high-velocity fall involves a distinct and measurable temporal pattern. Initially, there is a brief period of near-zero total acceleration—indicative of a free fall, a vehicle losing contact with the road, or a sudden airborne trajectory—followed immediately by an extreme, instantaneous deceleration spike upon the moment of impact, which is subsequently followed by an extended period of unnatural immobility.¹⁹

The raw sensor data outputted by the MEMS hardware contains excessive noise and requires rigorous algorithmic preprocessing to extract meaningful, distinguishable features before it can be classified.²³ The magnitude of linear acceleration, denoted as A_{mag} , and the magnitude of angular velocity, denoted as ω_{mag} , must be calculated continuously. This computational step simplifies the complex three-axis coordinate measurements into single scalar parameters that describe the overall kinematic force and rotational violence experienced by the user's device.²³

$$A_{mag} = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\omega_{mag} = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$$

While rudimentary detection systems have attempted to rely on simple static thresholding—for

instance, triggering an alert if tilt angles exceed 80 degrees on the X-axis and 70 degrees on the Y-axis, combined with vibration levels surpassing 4000 mV/G or sudden speed drops of over 20 km/h within three seconds—these methods are notoriously brittle.²³ Relying solely on static thresholds invariably results in unacceptably high false-positive rates, as the system struggles to differentiate a catastrophic vehicular collision from a user benignly dropping their phone onto a hard surface or braking sharply in heavy traffic. High false-positive rates in emergency systems lead to telecommunicator burnout and overwhelm Emergency Communications Centers (ECCs) with low-acuity events, pulling focus from genuine, life-and-death true emergencies.²⁴

To achieve the "Reliability" mandated by the hackathon judging criteria, the continuous stream of sensor data must be fed into a trained machine learning model deployed directly on the mobile device, ensuring low latency and eliminating the need for constant cloud connectivity. Empirical studies indicate that the duration of a typical crash or fall ranges from 1 to 3 seconds, with the most abrupt and defining physical changes occurring at the immediate beginning and ending of the kinetic event.²⁰

The data processing pipeline begins with data segmentation, where the application maintains a rolling, overlapping sliding window of the last 3 to 5 seconds of sensor data.²⁰ Feature extraction algorithms then process this window to generate a comprehensive feature vector. For example, data extracted from the X, Y, and Z axes can be integrated into a 1x44-dimensional row vector encompassing statistical features such as the slope of acceleration, variance, moving averages, standard deviation, and precise timestamps.²⁰

This high-dimensional feature vector is then input into the deployed classifier for continuous prediction.²⁰ The architectural choice of the machine learning model depends on the target device's computational constraints. Research has demonstrated that multibranch Convolutional Neural Networks (CNN) paired with Bidirectional Long Short-Term Memory (BiLSTM) networks excel at processing this specific type of time-series data, as the LSTM architecture is inherently designed to remember long-term dependencies and predict sequential behavioral anomalies.²¹ Alternative approaches, such as Hidden Markov Models (HMM) and Coupled HMM (CHMM), have also been successfully utilized for predicting driver behavior and crash dynamics based on sensor telemetry.²¹ For maximum compatibility across lower-tier Android devices, an implementation utilizing a Random Forest classifier or a lightweight TensorFlow Lite neural network provides excellent predictive accuracy while maintaining a negligible footprint on battery life and CPU utilization.²⁶

Upon deployment, when the model classifies an incoming sensor window as a "Crash" with a high confidence probability threshold, it initiates a fail-safe protocol. The application immediately triggers a loud auditory alarm and a high-contrast visual countdown timer (e.g., 10 seconds) on the device screen. If the user is uninjured or the event was a false positive, they can simply tap to cancel the countdown. If the countdown expires without intervention, the system assumes the user is incapacitated and automatically executes the core RoadSOS protocol: capturing the precise GPS coordinates, activating the offline-first dispatch sequence, and silently alerting nearby first responders and predefined emergency contacts.¹⁷

Geospatial Intelligence and Open Data Architectures

The fundamental utility of the RoadSOS application lies in its capacity to instantaneously fetch, rank, and display nearby emergency services relative to the user's current geographic location. This functionality requires a highly sophisticated geospatial intelligence framework capable of processing complex spatial queries with minimal latency. Given the hackathon's stringent evaluation criteria focusing on the "Number of contacts fetched" and "Data accuracy," relying on a single, proprietary mapping provider is a strategic vulnerability.²

OpenStreetMap (OSM) serves as the ideal foundational layer for this module. Often described as the "Wikipedia of maps," OSM is a collaborative, open-source global project that provides freely accessible, highly detailed spatial data.²⁹ The data structure of OSM is fundamentally different from traditional relational databases; it relies on three basic geometric elements: nodes (defining specific points with latitude and longitude), ways (ordered lists of nodes defining linear features like roads or area boundaries), and relations (groups of elements defining logical relationships, such as multipolygon buildings or bus routes).³⁰ Every element is heavily annotated with tags, consisting of key-value pairs that specify the precise characteristics of the object, such as `amenity=hospital`, `amenity=police`, or `emergency=ambulance_station`.³⁰

To interact with this massive global dataset, RoadSOS must leverage the Overpass API, a powerful, read-only database engine written in C++ that is specifically optimized for data consumers requiring selected parts of the OSM map based on complex search criteria like location, proximity, and tag properties.³² The application will utilize the Overpass Query Language (Overpass QL) to execute spatial bounding box queries.³³

When a crash is detected, the mobile application determines its current GPS coordinates and mathematically constructs a geographic bounding box (e.g., a 10-kilometer operational radius). The application then formats a specialized Overpass QL script. A highly optimized query designed to fetch all relevant emergency infrastructure simultaneously while preventing memory timeouts resembles the following syntax³¹:

Code snippet

```

[
  node[amenity=hospital] (around:10000, {latitude}, {longitude});
  node[amenity=police] (around:10000, {latitude}, {longitude});
  node[emergency=ambulance_station] (around:10000, {latitude}, {longitude});
];
```


This query instructs the Overpass API to search within a 10,000-meter radius of the incident, retrieve all nodes and ways matching the critical emergency tags, calculate the geometric center for complex multipolygon structures, and output the resultant dataset in a lightweight JSON format sorted by quadtile index (qt) for faster rendering.³⁰ Within the backend processing layer, written in Python, the requests library executes the HTTP POST to the Overpass interpreter, and the JSON response is parsed into structured formats using libraries like pandas and json_normalize, allowing the application to extract facility names, exact coordinates, and critically, the contact numbers stored in the phone=* tags.³¹ However, crowdsourced data, while expansive, can occasionally suffer from incomplete tagging. To ensure unassailable data accuracy and maximize the contact retrieval rate, RoadSOS must employ a dual-layer querying architecture by augmenting OSM data with authoritative, state-level Geographic Information System (GIS) APIs and Open Data portals. The Government of India and various state administrations have made significant strides in digital infrastructure.³⁹ For example, the state of Madhya Pradesh revolutionized its emergency response by instituting the 'Dial 100' project, establishing a centralized command center integrated with Hexagon Safety & Infrastructure's public safety software.⁴⁰ Furthermore, the Indore GIS platform provides a comprehensive digital system for collecting and analyzing geographic data, exposing spatial information regarding government hospitals, private clinics, police stations, and road networks on a unified digital map platform.³⁹ By establishing API connections to regional endpoints like the Indore GIS system or the U.S. Geological Survey's National Geospatial Data Asset (NGDA) for global scalability, RoadSOS cross-verifies the OpenStreetMap output.³⁹ If the OSM query returns a hospital node but lacks an associated phone number, the system automatically falls back to regional open-data directories, querying helplines such as 108 for ambulance services or 100 for police dispatch in Madhya Pradesh.⁴¹ This federated, multi-source data synthesis ensures that the first responders listed on the user's screen are highly accurate, directly satisfying the hackathon's "Reliability and data accuracy" criteria.²

Resilient Offline-First Architecture and Hardware Fallback

One of the most technically demanding evaluation criteria for the RoadSOS track is "Offline functionality".² The reality of vehicular travel dictates that accidents frequently occur on remote highways, traversing mountainous terrain, or during extreme weather events and natural disasters where cellular data networks (4G/5G) are entirely unavailable, overloaded, or physically compromised.⁴⁴ A conventional mobile application architected with a network-first or cache-first dependency paradigm becomes utterly paralyzed in these scenarios, presenting a blank screen precisely when the user's life depends on it.⁴⁶

To overcome this, RoadSOS must be engineered with an uncompromising offline-first architecture. This design philosophy inverts traditional software assumptions; rather than

treating disconnected operation as a fallback or edge case, offline-first design treats the disconnected state as the primary operational mode, with active internet connectivity acting merely as an opportunistic enhancement for data synchronization.⁴⁵ Research within disaster response technology indicates that mobile applications designed with robust, native offline capabilities can reduce emergency response times by up to 30 percent, a margin that translates directly into lives saved during the critical early minutes of a crisis.⁴⁵

Architectural Approach	Connectivity Dependency	User Experience Characteristics	Suitability for Emergency Response
Network-First	High	Application breaks completely without an active connection; infinite loading spinners.	Unacceptable; inherently dangerous for SOS tools.
Cache-First	Medium	Displays stale data; severely limited functional scope during offline periods.	Suboptimal; risk of routing to closed facilities.
Offline-First	Low	Instantaneous access to pre-loaded databases; functions identically in all network states.	Mandatory; ensures uninterrupted critical functionality.

The foundational layer of this offline-first module is a high-performance local database embedded directly within the mobile application. While modern alternatives like Realm or Core Data exist, SQLite remains the industry standard—the universally supported, battle-tested engine capable of storing vast amounts of relational data locally.⁴⁷ Upon initial installation, or whenever the application detects an unmetered Wi-Fi connection, RoadSOS proactively queries the Overpass API and government GIS endpoints to download the emergency infrastructure data (hospitals, trauma centers, police stations) corresponding to the user's home state and frequently traveled regional corridors.⁴⁸ This data is persistently stored in the local SQLite database. In the event of a crash, the application bypasses HTTP requests entirely, instantly querying the local SQLite tables to retrieve the nearest contacts based on the last

known GPS coordinates provided by the smartphone's internal GNSS chip, ensuring zero latency.⁴⁶

Providing contact numbers is insufficient; users in unfamiliar, remote territory require visual spatial context. Storing traditional raster map images (pixels) for offline viewing consumes prohibitive amounts of device storage space.⁵⁰ Therefore, RoadSOS must utilize packaged vector tiles, specifically adopting the MBTiles file format.⁵¹ MBTiles is an open specification based on a SQLite database containing Mapbox Vector Tiles (MVT) encoded in Protocolbuffer Binary Format (PBF).⁵¹ Vector data is dramatically more efficient than raster images, though it requires complex pre-calculation to filter and generalize geometries for different zoom levels.⁵⁰ Recent advancements in the Mapbox Maps SDK for iOS and Android support profound offline map capabilities, allowing users to download highly compressed tile packs.⁵² The second version of Mapbox offline map tile packs requires up to 40% less storage space, permitting the application to store vastly larger geographical regions without exhausting the mobile device's memory.⁵³ To comply with Mapbox API limitations, which restrict cumulative downloads to a maximum of 750 unique map tile packs per user, RoadSOS must intelligently manage the tile region loading process.⁵⁴ To render these offline vector maps, the system utilizes four core local assets: the JSON style descriptor file, the Glyphs files for typography, the Sprite files for rendering map icons, and the core MBTiles database containing the vector geometry.⁵¹ Utilizing Mapbox or the open-source MapLibre SDK, RoadSOS can render fluid, full-detail maps, guiding a panicked user to the nearest cached trauma center even when deep within a cellular dead zone.⁵¹

However, retrieving offline data locally only solves half the problem; the system must also possess the capability to transmit the SOS distress signal outward when internet connectivity is severed. When the AI crash detection module activates, and the application's network monitor detects a failure to establish a TCP/IP connection (HTTP API failure), it must gracefully and automatically degrade to an SMS-based fallback protocol.⁵⁵

The application programmatically constructs a highly compressed SMS payload string containing the critical incident parameters: the user's precise latitude and longitude, calculated kinetic severity (from the accelerometer data), and any pre-registered medical information (such as blood type or severe allergies). This SMS is silently dispatched via the cellular carrier's basic voice/text network, which often remains functional even when high-bandwidth data channels fail.

On the receiving end, to establish a truly resilient infrastructure that does not rely on third-party cloud APIs, emergency dispatch centers can deploy physical GSM Modems or hardware SMS gateways (such as SMSEagle or custom-built Arduino/Raspberry Pi configurations) running on-premises.²⁸ These physical modems interface directly with the cellular network and are controlled via a standardized set of AT (Attention) commands.⁵⁶ A typical automated initialization sequence for the receiving server involves resetting the modem to factory defaults using AT&F, configuring the SMS Service Centre Address for the specific mobile network using AT+CSCA, and setting the modem to Text Mode via AT+CMGF=1 or Protocol Data Unit (PDU) mode via AT+CMGF=0.⁵⁶ The on-premises Linux server continuously

polls the modem, parses incoming SMS distress signals, and visualizes them on real-time ArcGIS dashboards for dispatchers, ensuring that the alert reaches the authorities regardless of internet outages.⁵⁵

Integrated Health Infrastructure and Dynamic Triage Protocols

A critical failure point in modern emergency response is the misallocation of trauma resources. The hackathon's "Reliability and Data Accuracy" criterion necessitates that RoadSOS does not merely route a victim blindly to the nearest medical annotation on a map, but intelligently to the *clinically appropriate* medical facility.² In developing nations, including India, an ambulance can take anywhere between 15 to 40 minutes to reach an incident site in urban areas, and significantly longer in rural topographies.⁵⁹ The National Highways Authority of India has deployed over 900 ambulances at 50-kilometer intervals along national routes, yet thousands of individuals are deprived of their best chance of survival due to de facto pre-hospital delays and suboptimal routing.⁶⁰

The concept of the "Golden Hour" has served as a foundational principle in trauma and emergency medicine for decades, emphasizing the extreme urgency of providing definitive surgical care within the first 60 minutes following an injury.¹⁸ Originally coined by R. Adams Cowley as the "resuscitative hour," this concept shaped traditional trauma system design.¹⁸ However, as clinical data collection has matured, modern empirical research questions the strict rigidity of this 60-minute threshold. Emerging evidence from observational studies indicates that exceeding a 60-minute out-of-hospital time does not universally translate to increased mortality.¹⁸ Rather, studies have demonstrated that transporting a severely injured patient to a facility with specialized trauma capabilities is often significantly more critical to survival than merely minimizing transport time to an underequipped local hospital.³ For instance, spatial access studies in New Delhi revealed that for the most severely injured and time-sensitive cases, actively bypassing the nearest basic hospital in favor of rapidly transferring the victim to one of the 7 designated Level-1 trauma centres substantially improved survival outcomes, despite the longer transit.³

Consequently, RoadSOS must implement a smart algorithmic triage system based on the American College of Surgeons Committee on Trauma (ACS-COT) Field Triage Decision Scheme.⁶² When an accident is reported via the application, the interface generates a rapid, dynamic decision tree for bystanders or uninjured passengers. This algorithm assesses critical physiological and mechanistic criteria⁶²:

1. **Physiological Abnormalities:** Does the victim exhibit a systolic blood pressure (SBP) less than 90 mmHg, respiratory distress, or an altered Glasgow Coma Scale (GCS) motor score?⁶²
2. **Mechanistic and Anatomic Criteria:** Did the crash involve high velocities, rollover dynamics, or significant vehicular intrusion (e.g., inches of crushed cabin space)?⁶³

If the algorithm determines that the victim meets the high-acuity criteria, RoadSOS dynamically filters its internal SQLite database, actively excluding basic outpatient clinics (amenity=clinic)

and general practitioners (amenity=doctors), and exclusively routes the user to verified, designated trauma centres and surgical hospitals.³¹ The application utilizes distance matrix APIs to calculate travel time arrays, generating the fastest route to definitive care.³

To guarantee that the medical facilities recommended by the algorithm are genuinely operational, verified, and capable of handling trauma, RoadSOS must integrate natively with the Ayushman Bharat Digital Mission (ABDM), the Government of India's flagship national digital health infrastructure initiative.⁶⁵ Emerging from the National Digital Health Blueprint (NDHB), the ABDM creates a shared, interoperable ecosystem designed to securely exchange health data nationwide via APIs.⁶⁷

Developers participating in the hackathon can register on the ABDM Sandbox portal to implement the core building blocks of this digital ecosystem.⁶⁶ RoadSOS will specifically leverage three critical integration modules⁶⁸:

- **Module 2 (HFR Integration):** The application will query the Health Facility Registry (HFR) via API to verify the exact address, operational status, and medical specialty of nearby hospitals.⁶⁸ This ensures the application relies on authoritative, real-time government data rather than potentially outdated OpenStreetMap tags, perfectly satisfying the "Data accuracy" requirement.²
- **Module 1 (HPR Integration):** The system can query the Healthcare Professionals Registry (HPR) to ascertain the availability of verified trauma surgeons and specialists at the destination facility.⁶⁸
- **Module 3 (ABHA & Health Records):** If the user has pre-configured the application, their unique 14-digit Ayushman Bharat Health Account (ABHA) ID is automatically transmitted to the receiving trauma center's Health Information Exchange & Consent Manager (HIE-CM) dashboard.⁶⁸ This allows the emergency room physicians to decrypt and review the patient's longitudinal medical history, blood type, and severe allergies packaged as FHIR (Fast Healthcare Interoperability Resources) bundles before the ambulance even arrives at the hospital bay.⁶⁹

Global Protocol Resolution and Cross-Border Routing

Road traffic, commercial freight logistics, and tourism are inherently transnational activities. Therefore, a primary evaluation criterion established by the IIT Madras hackathon jury is the "Information integration across countries".² An application designed strictly for the Indian subcontinent becomes entirely useless the moment a user crosses the border into Nepal, Bangladesh, or travels overseas. A truly resilient emergency platform requires dynamic geographic adaptability.

RoadSOS must incorporate a continuous geospatial geofencing engine that silently monitors the device's coordinates at the macroscopic, national level. When the application detects that the user has crossed an international border, it must dynamically update its internal routing tables to match the specific emergency protocols, nomenclature, and telecommunications codes of the host nation.

To achieve this, the architecture will maintain a comprehensive SQLite mapping table of global emergency telephone numbers, with a specific focus on the BIMSTEC (Bay of Bengal Initiative

for Multi-Sectoral Technical and Economic Cooperation) region and prominent global destinations.⁷⁰

Sovereign State	Primary Police	Emergency Ambulance	Fire & Rescue	General / Mobile Fallback	Notes & Regional Exceptions
India	100	102 / 108	101	112	ERSS-112 unifies multiple services nationally. ⁶⁰
Bangladesh	999	999	999	999	Dhaka Foreigner Desk: 01700-700 700. ⁷⁴
Nepal	100	102	101	112	Traffic Police direct line: 103. ⁷¹
Sri Lanka	119	110	110	112 / 119	Traffic Police direct line: 112-691-111. ⁷⁰
Thailand	191	1669	199	112	Tourist Police: 1155; Bangkok Ambulance: 1646. ⁷¹
Taiwan (ROC)	110	119	119	112	Mobile dial 112, redirect 0 for police, 9 for fire. ⁷¹

The system must account for intricate localized nuances beyond simple three-digit numbers. For instance, in Bangladesh, the government officially launched the toll-free 999 emergency

helpline managed by the police force; however, the Dhaka Metropolitan Police maintains specialized email and mobile contacts (e.g., 01700-700700) specifically engineered to assist foreign nationals, which the app should prioritize if the user's registered home country is outside Bangladesh.⁷⁴ Similarly, in Thailand, while 1669 remains the standard national medical number, a foreign tourist involved in a vehicular collision may be better served by the multilingual Tourist Police at 1155, or the specialized Bangkok Metro traffic control center at 1197.⁷¹

From a software engineering perspective, the Python backend or the on-device logic will utilize reverse geocoding libraries or raw boundary queries against the OSM dataset `admin_level=2` tagged with the ISO3166-1 standard to ascertain the current country code.³⁷ Upon detecting an ISO string change (e.g., moving from `ISO3166-1="IN"` to `ISO3166-1="NP"`), the application seamlessly updates the primary UI action buttons, rewriting the dialer intents to call 100 instead of 112, and formatting Overpass queries to prioritize host-nation amenities. This automated, zero-configuration global adaptability guarantees maximum scoring on the cross-country integration metric.

Strategic Deliverable Synthesis and Hackathon Submission

Transitioning this expansive, multi-layered architectural blueprint into a successful Stage 1 submission for the National Road Safety Hackathon 2026 requires meticulous adherence to the structural competition guidelines and an acute awareness of the evaluative focus of the CoERS jury.⁵

The codebase itself should ideally be developed using a highly portable, cross-platform mobile framework such as Google's Flutter or Meta's React Native. This ensures operability and identical user experiences across both Android and iOS operating systems, maximizing the total addressable user base. The codebase must cleanly modularize the Edge AI inference engine (importing the TensorFlow Lite models for crash detection), the local database layer (initializing the SQLite schemas for offline caching), and the mapping interface (integrating the Mapbox or MapLibre SDKs for MBTiles vector rendering).²⁷ The backend analytical components and API routers should be engineered in a robust language like Python, utilizing the requests library to interface concurrently with the Overpass API endpoint (<http://overpass-api.de/api/interpreter>) and the various ABDM gateway endpoints.³⁷

A mandatory component of the Stage 1 submission protocol is the compilation of a Word document rigorously detailing programmatic assumptions and software packages.⁵ To satisfy this requirement comprehensively, the document must state the operational parameters of the system.

- **Critical Assumptions to Document:** The application assumes the availability of a relatively unobstructed line-of-sight to GNSS/GPS satellite constellations for accurate coordinate generation; it assumes a statistically acceptable margin of error within the OpenStreetMap crowdsourced data prior to ABDM cross-verification; and it assumes the presence of rudimentary 2G cellular network capabilities for the execution of the SMS GSM Modem fallback protocol during 4G/5G data outages.

- **Software Packages Manifest:** The document must explicitly list all dependencies to prove the viability of the code. This includes Python packages such as osm2geojson for parsing the complex geometric relations of map data into GeoJSON format ³⁸, scikit-learn or tensorflow-lite for the implementation and porting of the machine learning classifiers ¹⁹, and the precise versioning of the Mapbox SDK utilized to comply with the 750 tile pack limitation. ⁵²

Finally, the required seven-slide presentation deck must distill this immense architectural complexity into a concise, compelling narrative designed for a live jury presentation.⁵ An optimized, highly strategic flow for this deck is as follows:

1. **Slide 1: Welcome & Executive Summary** - Formal introduction of the team, displaying the RoadSOS application interface, and stating the core value proposition.
2. **Slide 2: The Macro Problem Space** - Quantifying the 1.7 lakh annual fatalities, the economic drain of traffic accidents, and the critical pre-hospital delays contributing to the failure of the "Golden Hour".¹
3. **Slide 3: Systems Architecture Diagram** - A high-level visual schematic showcasing the continuous data flow between the mobile application, the Overpass API, the state GIS portals (like MP Dial 100), and the ABDM health registries.⁴⁰
4. **Slide 4: AI Crash Detection (Innovation Focus)** - Detailing the physics of the accelerometer/gyroscope sensor fusion, the data segmentation process, and the superior accuracy of the edge-based CNN-BiLSTM or Random Forest models.²¹
5. **Slide 5: Unyielding Resilience (Offline-First Focus)** - Explaining the inversion of the cache-first model, detailing the MBTiles SQLite vector mapping strategy, and diagramming the SMS hardware gateway utilizing AT commands.⁴⁹
6. **Slide 6: Intelligent Triage & Global Scale** - Showcasing the dynamic ACS-COT physiological decision tree, the integration with ABHA for instant medical record transfer, and the automated ISO 3166-1 geographic routing tables for cross-border compatibility.³⁷
7. **Slide 7: Thank You & Alignment Vision** - Concluding by explicitly aligning the RoadSOS architecture with CoERS's mission of Data-Driven Hyperlocal Interventions (DDHI) and demonstrating how the tool integrates seamlessly with platforms like Sanjaya and RATH to foster a permanent, data-driven safety culture in India.¹

The National Road Safety Hackathon 2026 presents an unparalleled opportunity to engineer a paradigm shift in emergency response dynamics. A successful implementation of RoadSOS must deliberately abandon the triviality of static contact directories. By embedding machine learning models directly at the edge for instantaneous kinetic crash detection, synthesizing highly diverse datasets from global OpenStreetMap nodes and the centralized Ayushman Bharat Digital Mission, and architecting an uncompromisingly resilient offline-first communication framework, systems architects can create a civic tool of profound utility. This comprehensive, multi-layered approach strictly satisfies the exhaustive evaluation criteria established by IIT Madras. More critically, it addresses the systemic, root-cause deficiencies in pre-hospital emergency care. By ensuring that victims are routed not merely with speed, but intelligently to the appropriately equipped surgical trauma centres, and by guaranteeing system

operability regardless of severe geographical isolation or catastrophic network failure, RoadSOS possesses the tangible capability to drastically reduce the morbidity and mortality associated with vehicular accidents. Integrating this technological suite within the broader, sophisticated governance frameworks of Sanjaya and RATH will ultimately realize the institutional vision of a proactive, data-driven, and highly empathetic road safety ecosystem.

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